HW4 Hypothesis

A(i): I hypothesize that when the coefficient α is positive, the oscillation will decrease in amplitude and approach to 0. For underdamped cases, the signal will oscillating and decreasing in amplitude. For critically damped and overdamped cases, the signal will return to equilibrium without oscillate. When $\alpha = 0$, the oscillation will maintain the current form since there is no dampening. When $\alpha < 0$, the amplitude will increase because the signal is gaining energy.

A(ii): The Van der Pol oscillator will reach stable states regardless the initial condition. The steady state amplitude should be the same no matter different starting points given the regularization term $1 - x(t)^2$. The larger α should have increase the oscillation period because the stronger nonlinear effects will make the waveform less sinusoidal and provide slower transition.

(iii): Based on the McSharry-Clifford model, I hypothesize that changing the amplitude a_i and width b_i will change the corresponding ECG components. The amplitude parameters a_Q, a_R, a_S and width parameters b_Q, b_R, b_S will affect the QRS complex. The a_T and b_T will affect the T waves. The larger amplitude parameter will produce stronger waveform and larger width parameter will produce broader waves.